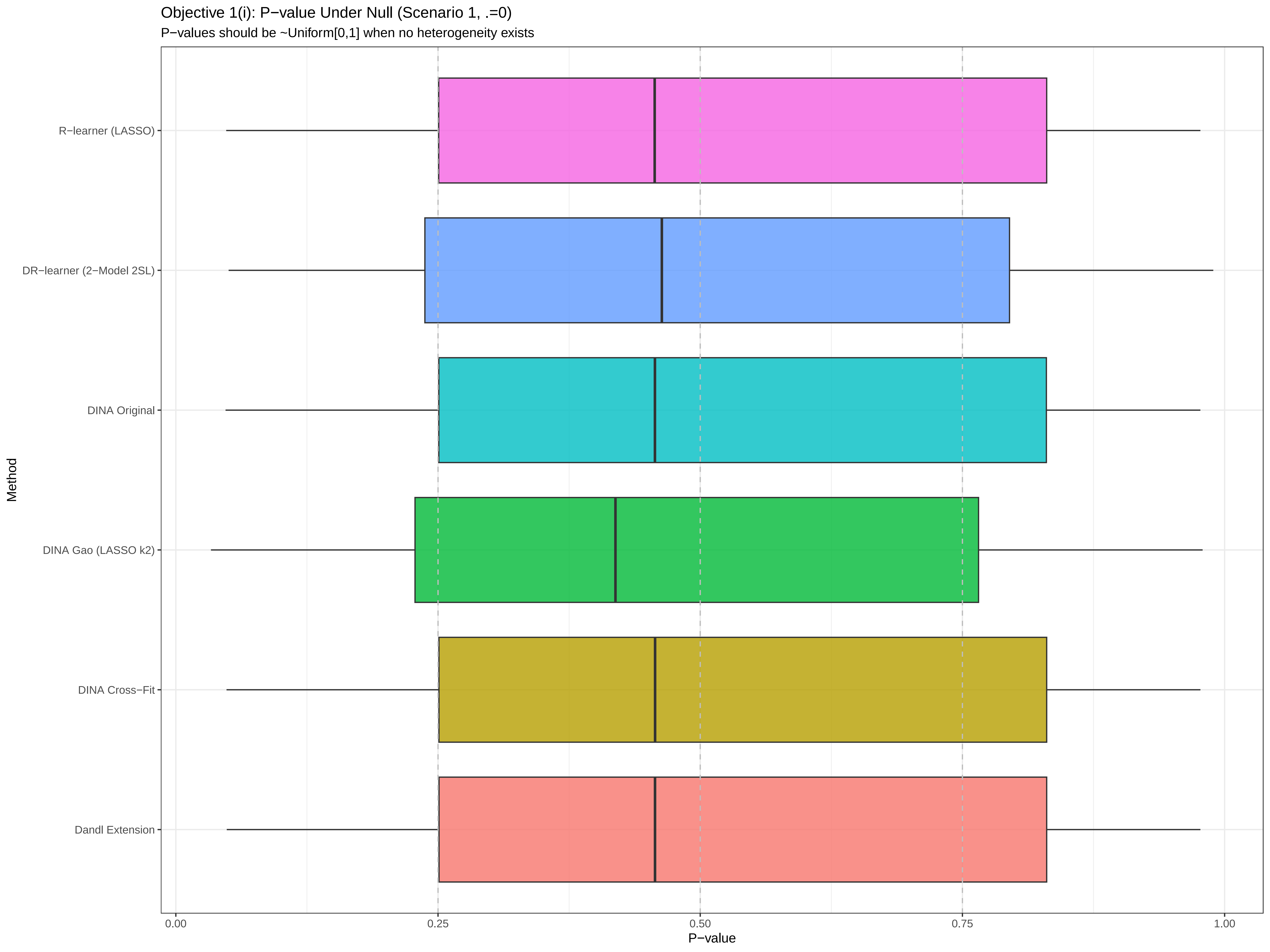
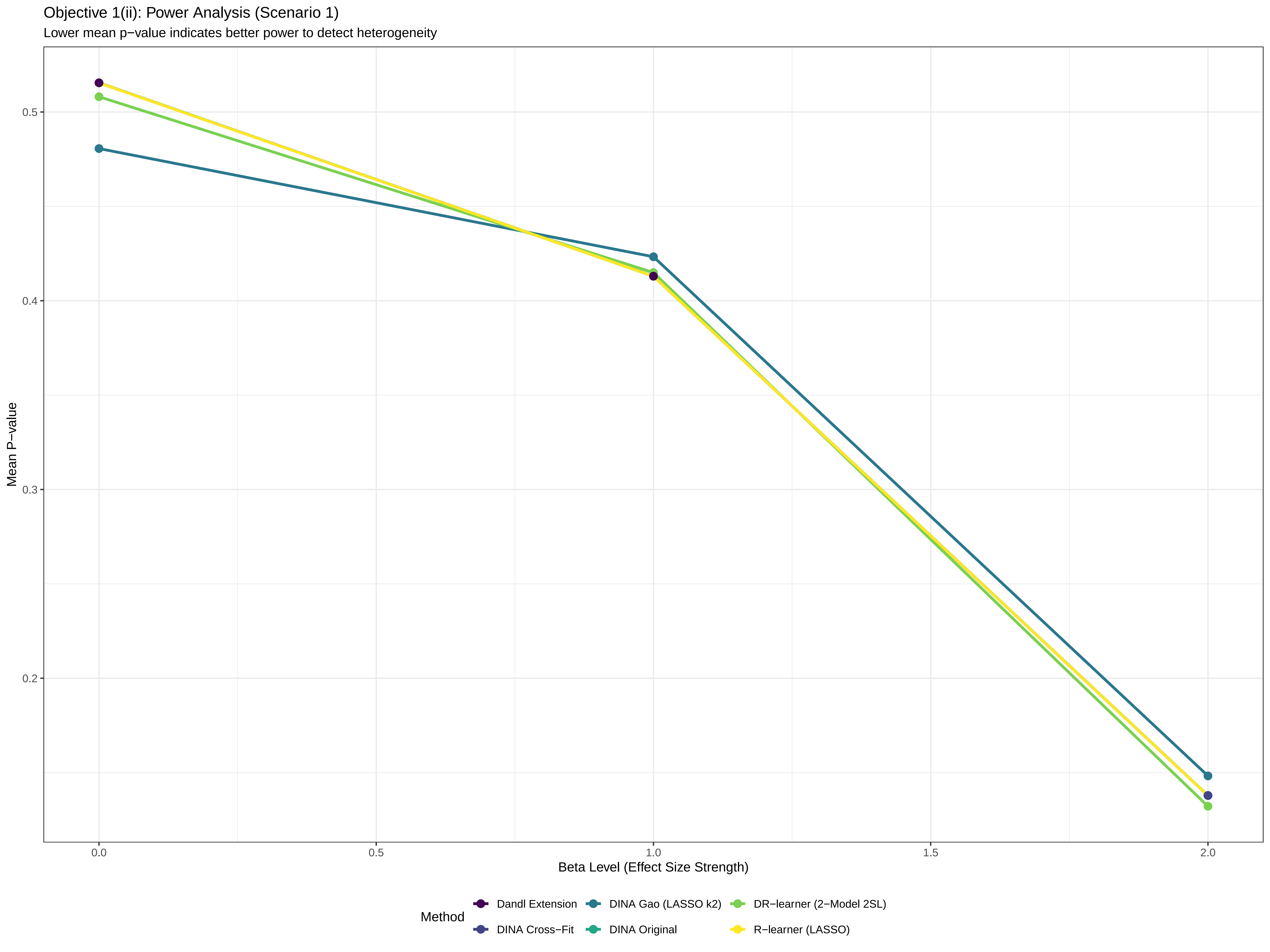
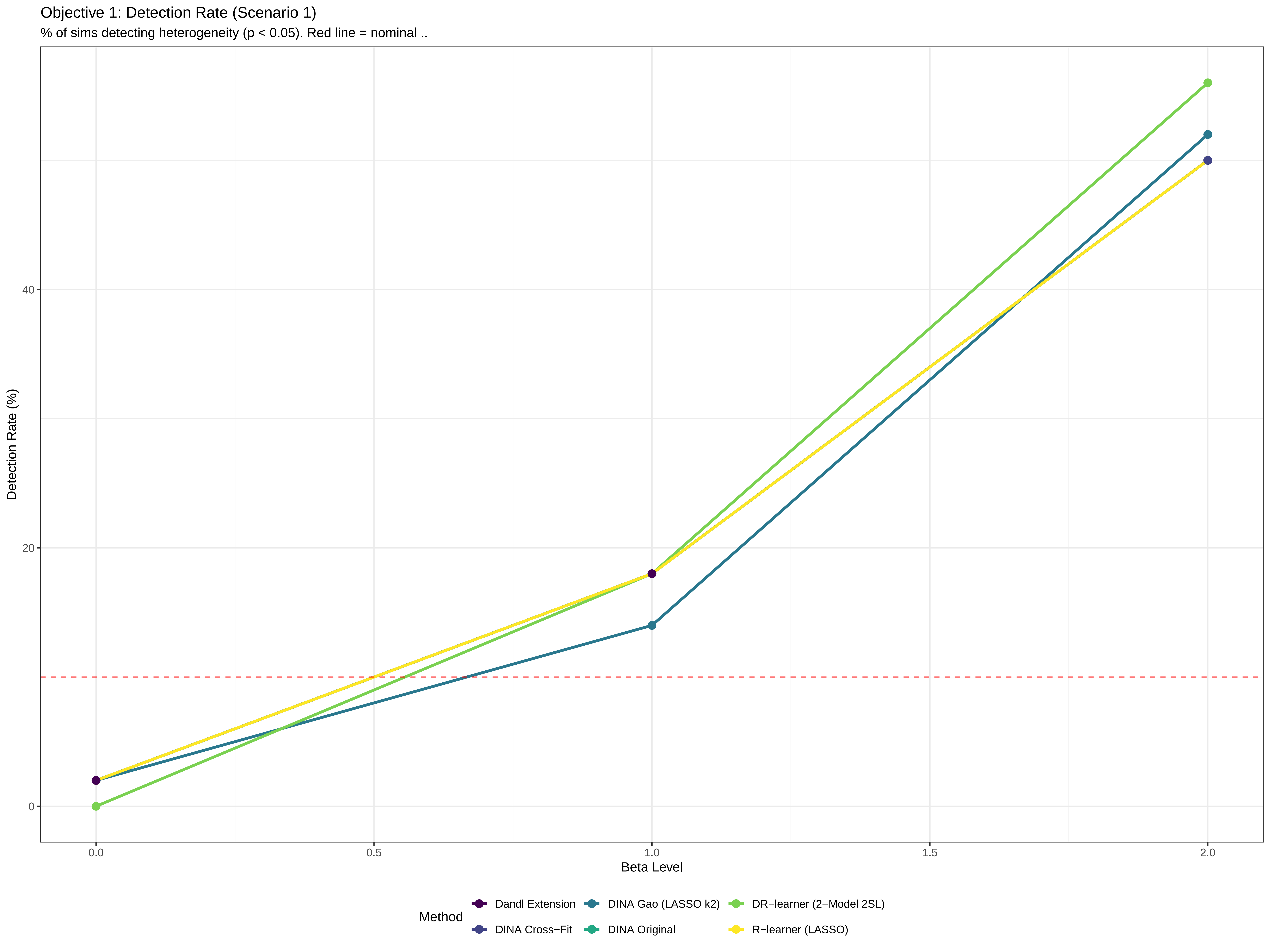

Objective 1(i): P−value Under Null (Scenario 1, .=0)
P−values should be ~Uniform[0,1] when no heterogeneity exists
R−learner (LASSO)
DR−learner (2−Model 2SL)
DINA Original
DINA Gao (LASSO k2)
DINA Cross−Fit
Dandl Extension
Method
0.00
0.25
0.50
0.75
1.00
P−value



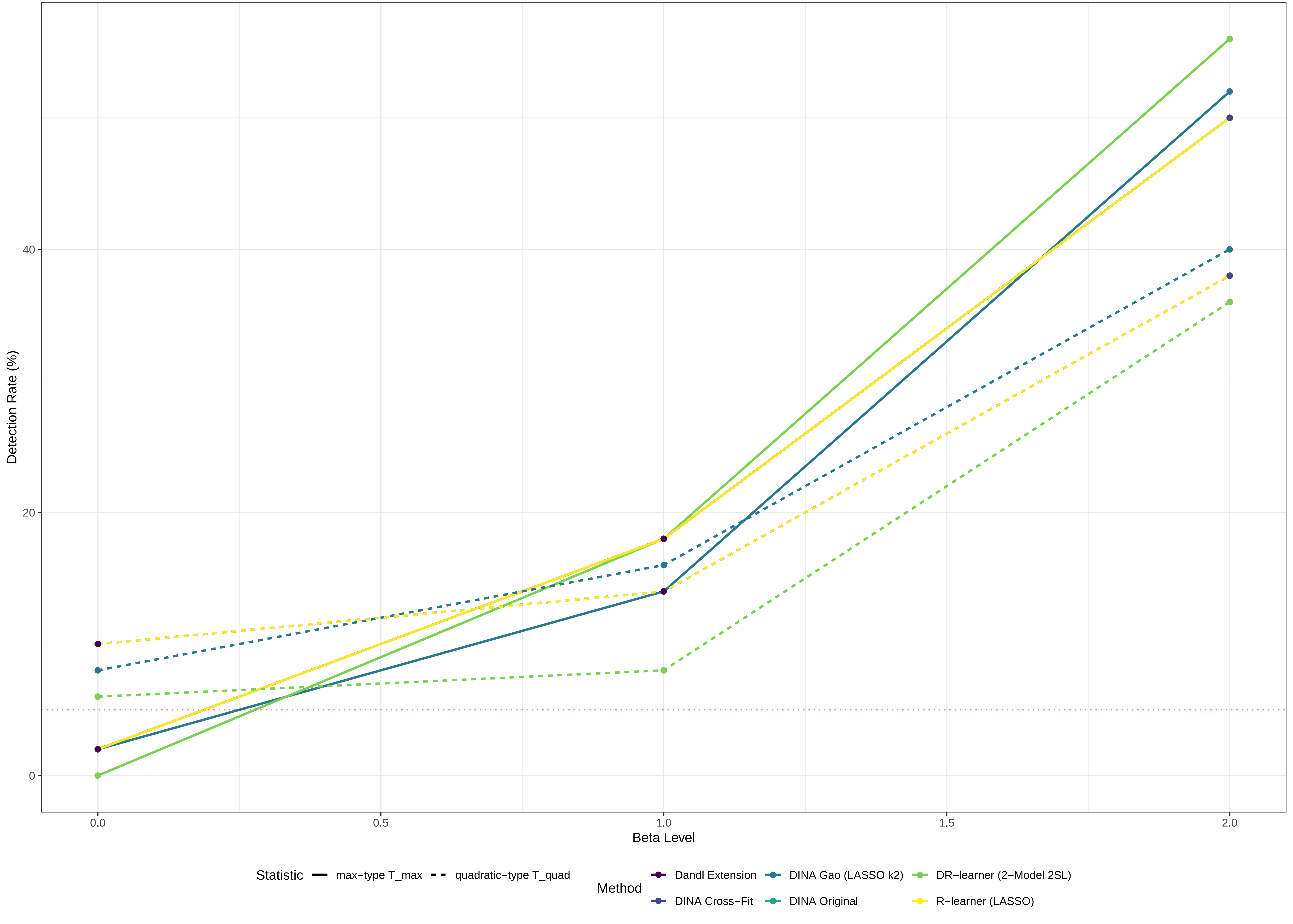
Objective 1(ii): Power Analysis (Scenario 1)
Lower mean p−value indicates better power to detect heterogeneity
Mean P−value
0.5
0.4
0.3
0.2
0.0
0.5
1.0
1.5
2.0
Beta Level (Effect Size Strength)
Method
Dandl Extension
DINA Cross−Fit
DINA Gao (LASSO k2)
DINA Original
DR−learner (2−Model 2SL)
R−learner (LASSO)

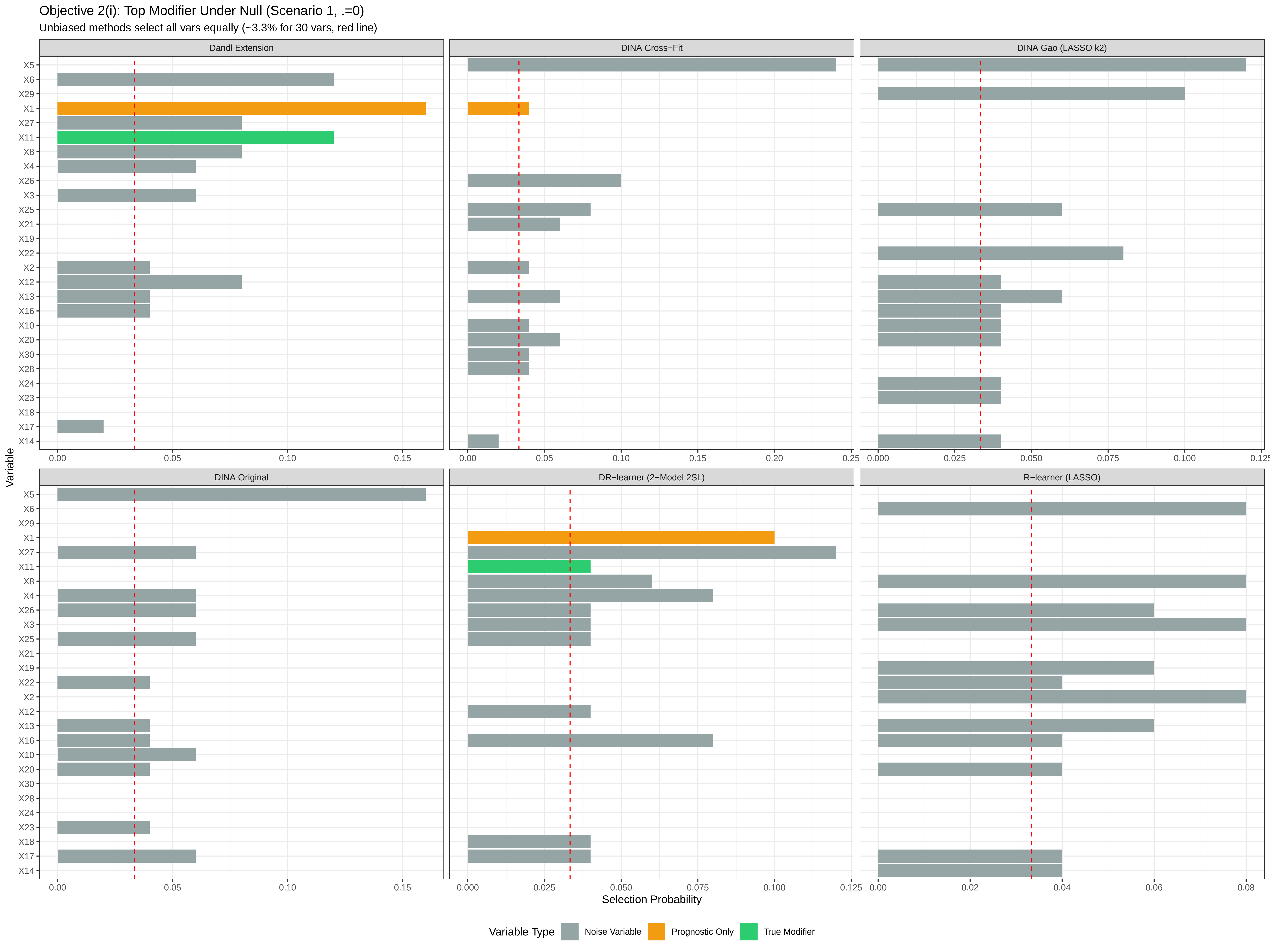


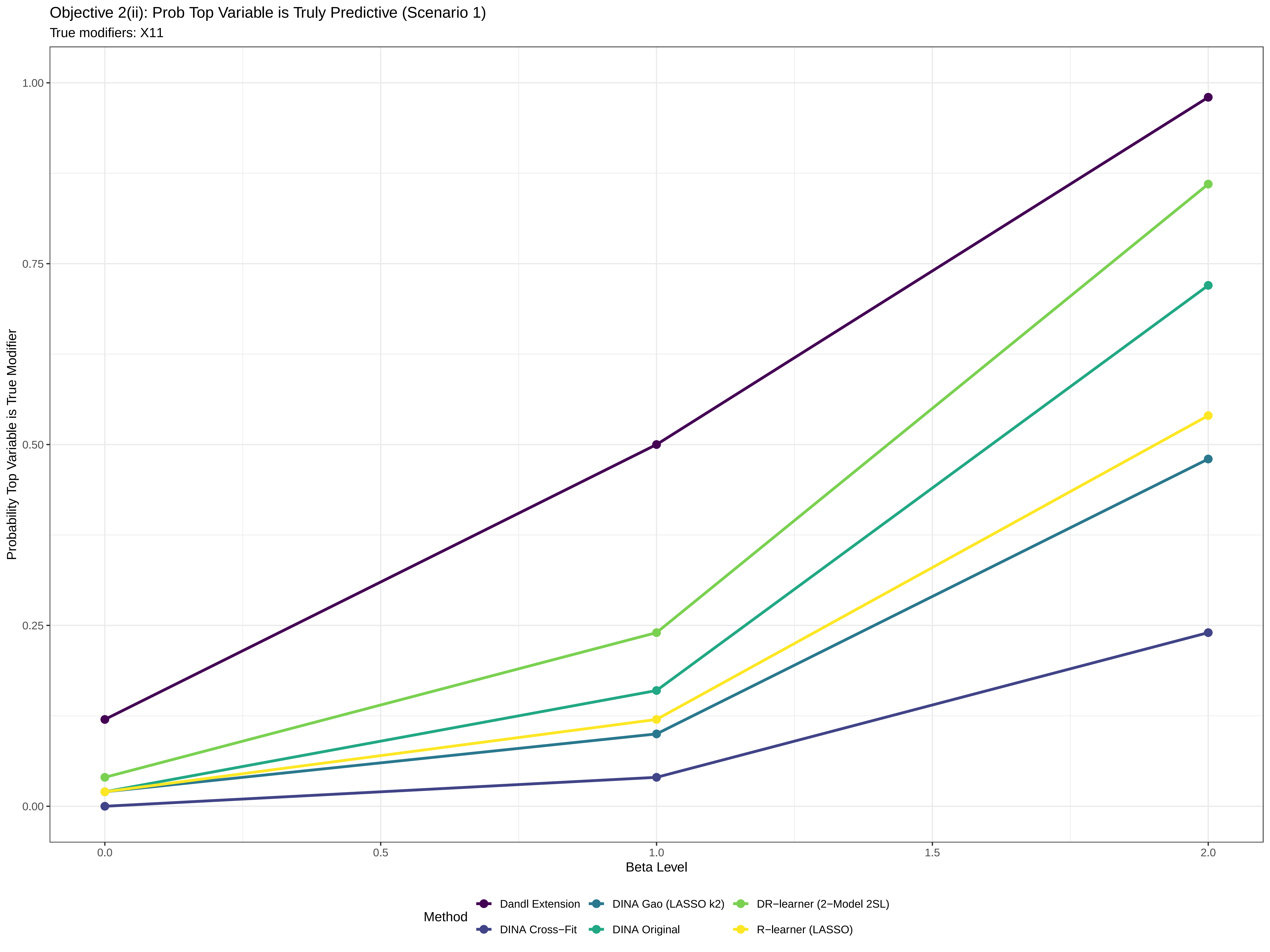
Objective 1: Detection Rate (Scenario 1)
% of sims detecting heterogeneity (p < 0.05). Red line = nominal ..
Detection Rate (%)
0
20
40
0.0
0.5
1.0
1.5
2.0
Beta Level
Method
Dandl Extension
DINA Cross−Fit
DINA Gao (LASSO k2)
DINA Original
DR−learner (2−Model 2SL)
R−learner (LASSO)

Objective 1: Max− vs Quadratic−type Power (Scenario 1)

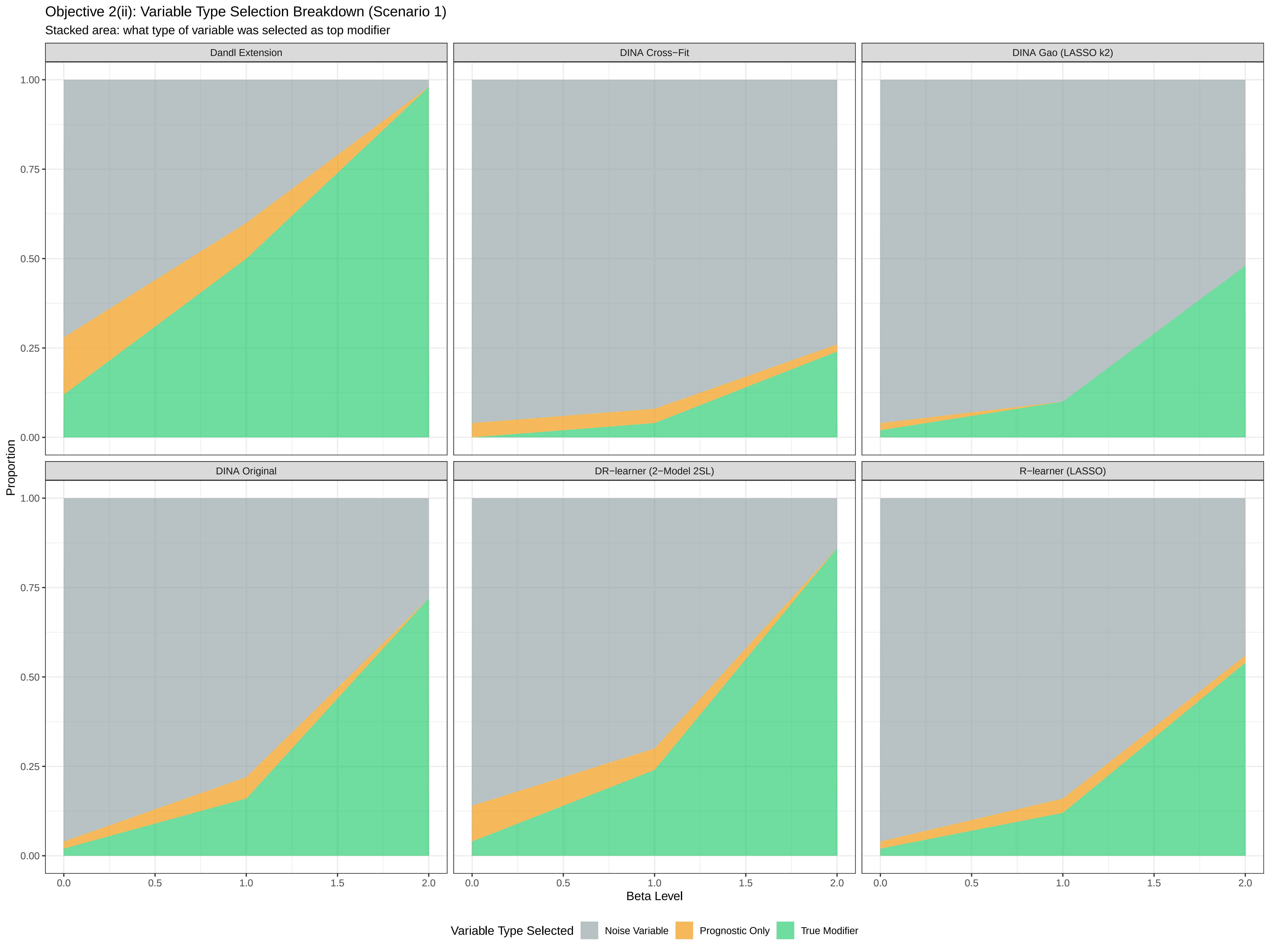
Permutation test on each method's pseudo−outcome. Red dotted = nominal . = 5%.



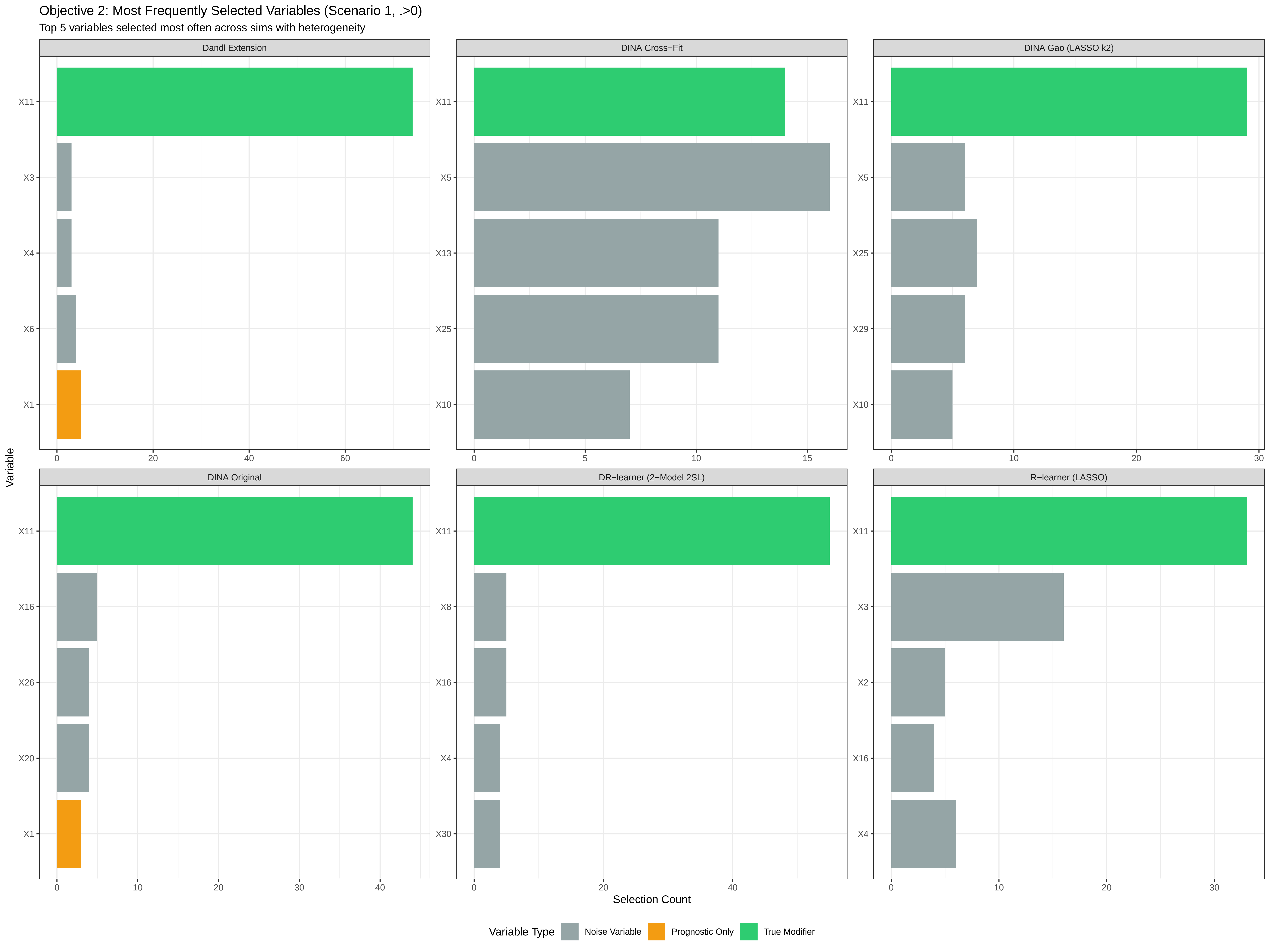

Objective 2(i): Top Modifier Under Null (Scenario 1, .=0)
Unbiased methods select all vars equally (~3.3% for 30 vars, red line)
Dandl Extension
DINA Cross−Fit
DINA Gao (LASSO k2)
DINA Original
DR−learner (2−Model 2SL)
R−learner (LASSO)
Variable
Selection Probability
Variable Type
Noise Variable
Prognostic Only
True Modifier

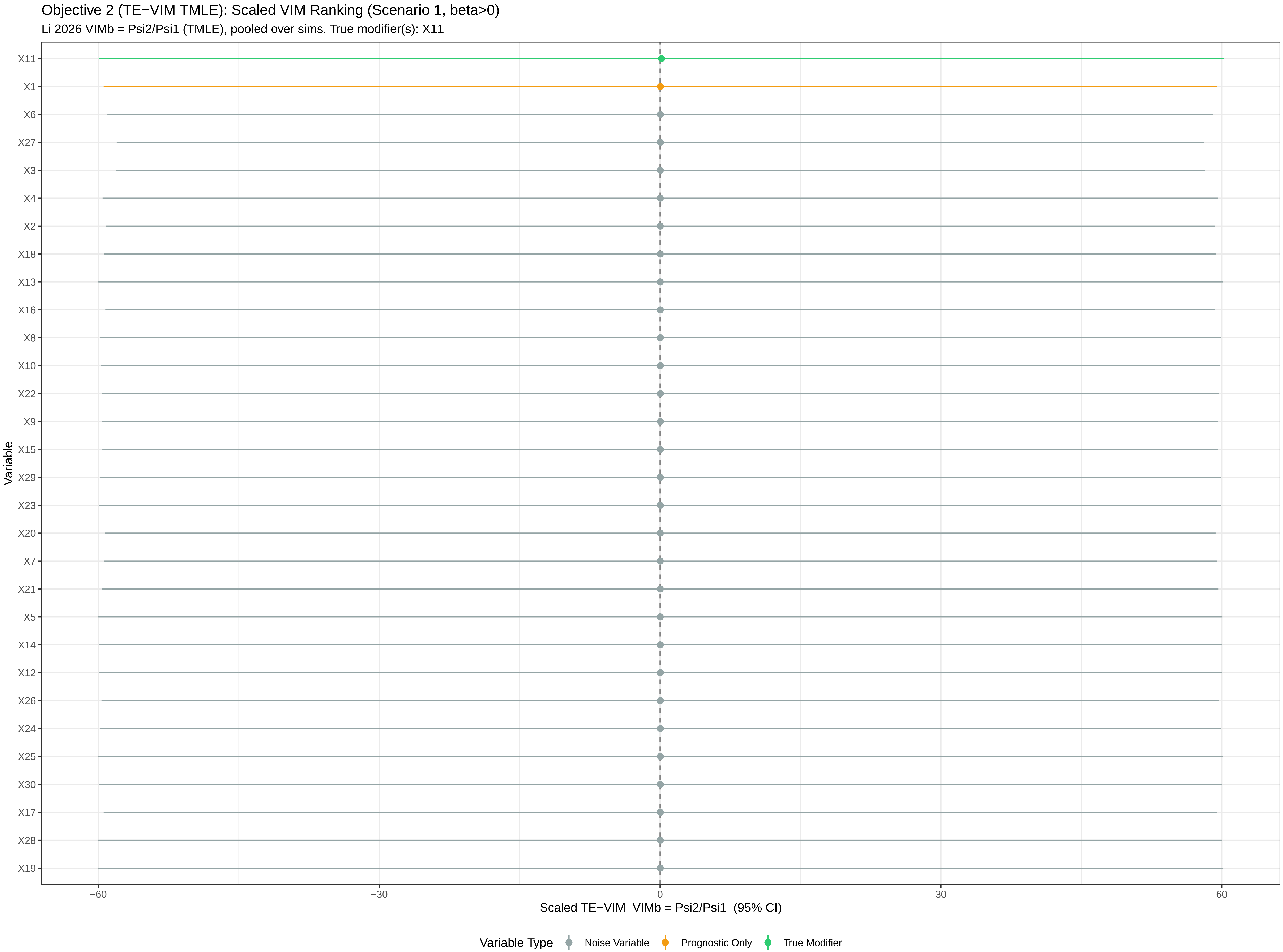


Objective 2(ii): Prob Top Variable is Truly Predictive (Scenario 1)
True modifiers: X11
Probability Top Variable is True Modifier
1.00
0.75
0.50
0.25
0.00
0.0
0.5
1.0
1.5
2.0
Beta Level
Method
Dandl Extension
DINA Cross−Fit
DINA Gao (LASSO k2)
DINA Original
DR−learner (2−Model 2SL)
R−learner (LASSO)

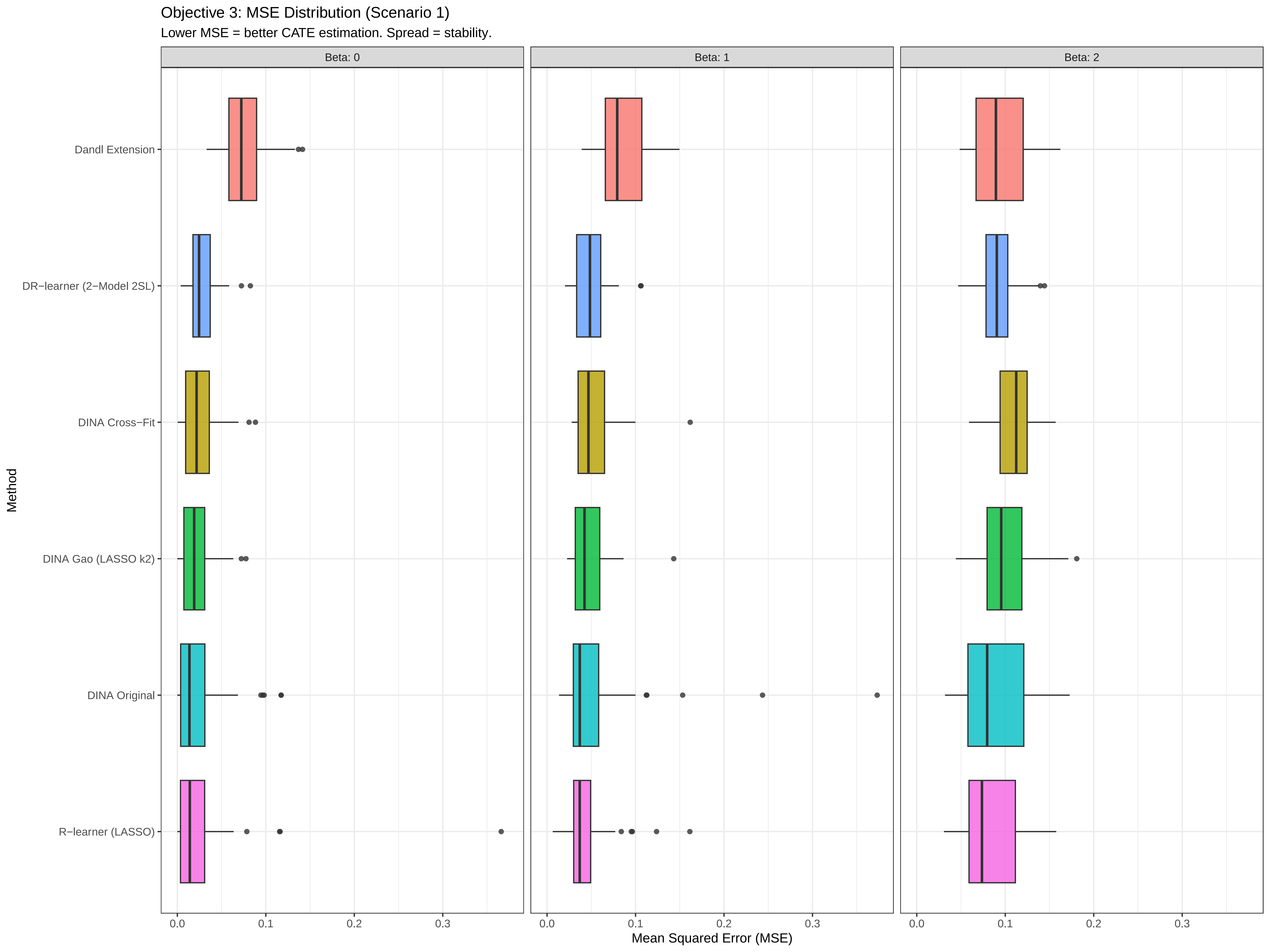


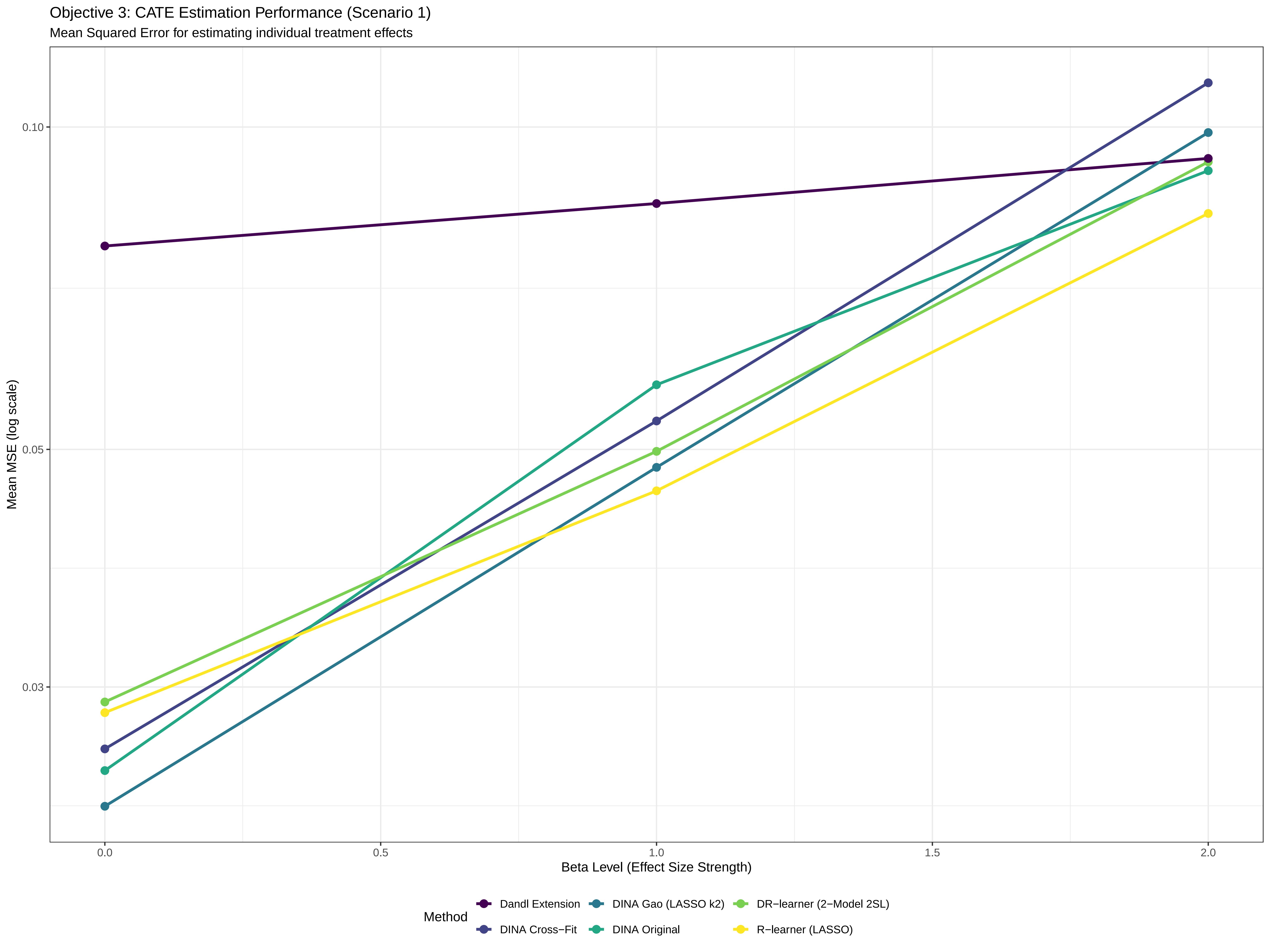
Objective 2(ii): Variable Type Selection Breakdown (Scenario 1)
Stacked area: what type of variable was selected as top modifier
Dandl Extension
DINA Cross−Fit
DINA Gao (LASSO k2)
DINA Original
DR−learner (2−Model 2SL)
R−learner (LASSO)
Proportion
1.00
0.75
0.50
0.25
0.00
0.0
0.5
1.0
1.5
2.0
Beta Level
Variable Type Selected
Noise Variable
Prognostic Only
True Modifier

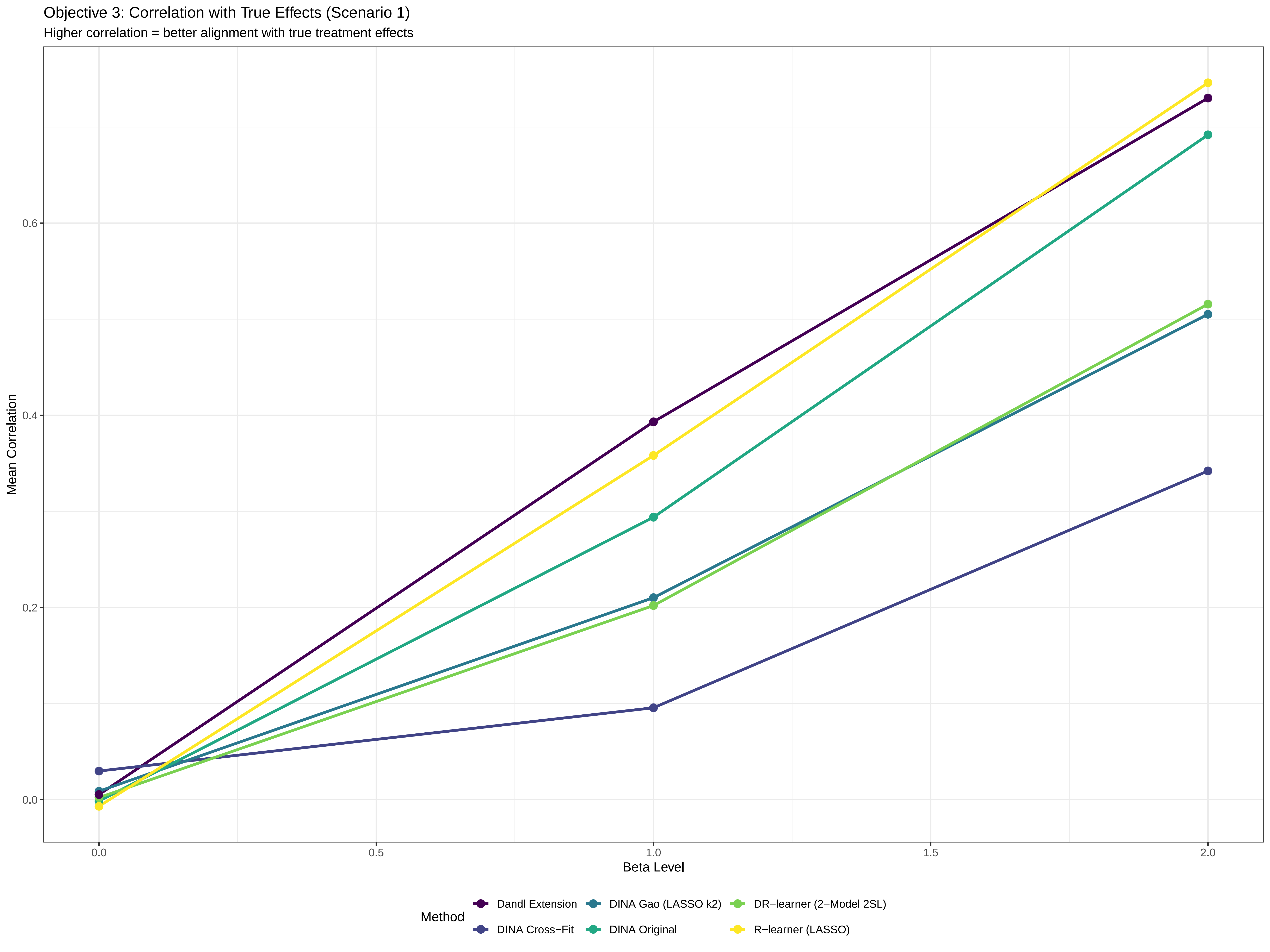

Objective 2: Most Frequently Selected Variables (Scenario 1, .>0)
Top 5 variables selected most often across sims with heterogeneity
Dandl Extension
DINA Cross−Fit
DINA Gao (LASSO k2)
DINA Original
DR−learner (2−Model 2SL)
R−learner (LASSO)
X11
X3
X4
X6
X1
X5
X13
X25
X10
X29
X16
X26
X20
X8
X30
X2
Variable
Selection Count
Variable Type
Noise Variable
Prognostic Only
True Modifier

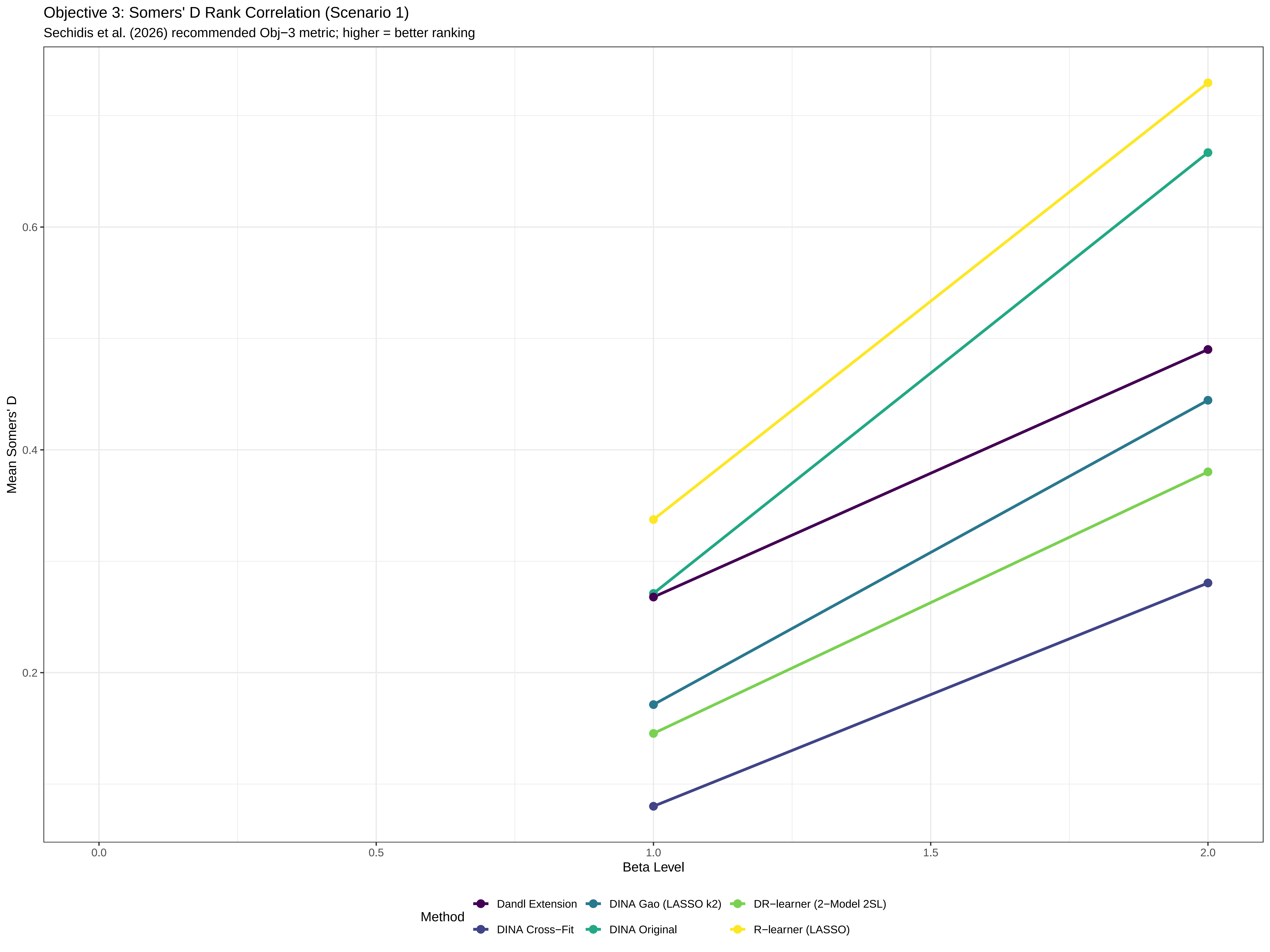


Objective 2 (TE−VIM TMLE): Scaled VIM Ranking (Scenario 1, beta>0)
Li 2026 VIMb = Psi2/Psi1 (TMLE), pooled over sims. True modifier(s): X11
Variable
X11
X1
X6
X27
X3
X4
X2
X18
X13
X16
X8
X10
X22
X9
X15
X29
X23
X20
X7
X21
X5
X14
X12
X26
X24
X25
X30
X17
X28
X19
−60
−30
0
30
60
Scaled TE−VIM VIMb = Psi2/Psi1 (95% CI)
Variable Type
Noise Variable
Prognostic Only
True Modifier


Objective 3: MSE Distribution (Scenario 1)
Lower MSE = better CATE estimation. Spread = stability.
Beta: 0
Beta: 1
Beta: 2
Dandl Extension
DR−learner (2−Model 2SL)
DINA Cross−Fit
DINA Gao (LASSO k2)
DINA Original
R−learner (LASSO)
Method
0.0
0.1
0.2
0.3
Mean Squared Error (MSE)


Objective 3: CATE Estimation Performance (Scenario 1)
Mean Squared Error for estimating individual treatment effects
Mean MSE (log scale)
0.10
0.05
0.03
0.0
0.5
1.0
1.5
2.0
Beta Level (Effect Size Strength)
Method
Dandl Extension
DINA Cross−Fit
DINA Gao (LASSO k2)
DINA Original
DR−learner (2−Model 2SL)
R−learner (LASSO)


Objective 3: Correlation with True Effects (Scenario 1)
Higher correlation = better alignment with true treatment effects
Mean Correlation
0.0
0.2
0.4
0.6
0.0
0.5
1.0
1.5
2.0
Beta Level
Method
Dandl Extension
DINA Cross−Fit
DINA Gao (LASSO k2)
DINA Original
DR−learner (2−Model 2SL)
R−learner (LASSO)


Objective 3: Somers' D Rank Correlation (Scenario 1)
Sechidis et al. (2026) recommended Obj−3 metric; higher = better ranking
Mean Somers' D
0.2
0.4
0.6
0.0
0.5
1.0
1.5
2.0
Beta Level
Method
Dandl Extension
DINA Cross−Fit
DINA Gao (LASSO k2)
DINA Original
DR−learner (2−Model 2SL)
R−learner (LASSO)

Summary Comparison (Scenario 1) − averaged over .>0
True Modifiers: X11

Method	Avg_Power	Avg_Detection	Avg_MSE	Avg_Correlation	Avg_Somers_D	Avg_True_Selection
DR−learner (2−Model 2SL)	0.726	37	0.071	0.359	0.263	55
R−learner (LASSO)	0.725	34	0.064	0.552	0.533	33
DINA Cross−Fit	0.725	34	0.082	0.219	0.180	14
Dandl Extension	0.725	34	0.089	0.562	0.379	74
DINA Original	0.725	34	0.074	0.493	0.469	44
DINA Gao (LASSO k2)	0.714	33	0.073	0.358	0.308	29